

Order Flow Imbalance and Price Impact in CME Ether Futures

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Abstract

We study Order Flow Imbalance (Cont, Kukanov, and Stoikov, 2014) on CME Ether futures from February 2021 to August 2026. On 28,711 out-of-sample dollar-volume bars, an OLS regression of bar mid-quote change on contemporaneous and one-bar lagged OFI with Newey-West HAC standard errors yields a contemporaneous coefficient $\hat{\beta}_0 = +0.215$ ($t = 58.1$, $R^2 = 0.30$), against $+0.300$ and $R^2 = 0.31$ in sample. The regression is contemporaneous, so it measures price impact rather than forecasting power: used on its own as a predictor, lagged OFI explains essentially none of the following bar's move ($R^2 < 0.001$ out of sample). OFI is a strong descriptor of price formation on this contract; it is not, by itself, a trading signal.

1 Introduction

Order Flow Imbalance (OFI) aggregates best-bid and best-offer quote arrivals into a signed measure of top-of-book pressure. On liquid US equities, contemporaneous OFI explains roughly two-thirds of mid-quote variation over ten-second intervals (Cont, Kukanov, and Stoikov, 2014). The cryptocurrency evidence is thinner: Silantyev (2019) finds trade-flow imbalance dominates short-horizon prices on BitMEX XBT/USD, and Bieganski and Ślepaczuk (2026) report OFI as the top SHAP feature in CatBoost ensembles on Binance Futures perpetuals. CME's Ether contract sits apart from both: it trades on a regulated venue, and its quoted spread, in basis points of price, is several times the equity tape's.

This paper estimates the contemporaneous price impact of OFI on CME ETH futures from February 2021 to August 2026. The construction is the level-1 (best bid / best offer) form of Cont et al. on dollar-volume bars built from TBBO (trade-sampled top-of-book) events (Lopez de Prado, 2018).

2 Data

The panel consists of CME-cleared TBBO data for the front-month ETH futures contract, covering February 2021 to August 2026. Bars never span a contract roll, and the mid-quote change across each roll is excluded from the regression.

Bars are formed on a one-minute grid: TBBO events are aggregated to minutes, and a bar closes at the first minute at which its cumulative dollar volume crosses a threshold set to one tenth of the median in-sample daily dollar volume and held fixed out of sample.

For each bar the following quantities are computed:

*Independent research conducted by the author; the views expressed are the author's own and not those of Imperial College London.

- **ofi**: signed level-1 order flow imbalance, following the construction of Cont et al. (2014). For each TBBO update i , let p_i^B, p_i^A denote the best-bid and best-ask quotes and q_i^B, q_i^A their sizes. The per-event OFI contribution is $e_i = e_i^B + e_i^A$, where

$$e_i^B = \begin{cases} +q_i^B & \text{if } p_i^B > p_{i-1}^B \\ -q_{i-1}^B & \text{if } p_i^B < p_{i-1}^B \\ q_i^B - q_{i-1}^B & \text{if } p_i^B = p_{i-1}^B \end{cases}, \quad e_i^A = \begin{cases} -q_i^A & \text{if } p_i^A < p_{i-1}^A \\ +q_{i-1}^A & \text{if } p_i^A > p_{i-1}^A \\ -(q_i^A - q_{i-1}^A) & \text{if } p_i^A = p_{i-1}^A. \end{cases}$$

The bar OFI is the unnormalised sum over all events in the bar, $O_t = \sum_{i \in \text{bar } t} e_i$.

- **mid.close**: the mid quote $(p^B + p^A)/2$ at the bar boundary.
- **d.mid**: the change in mid quote since the previous bar.

3 Price-impact regression

Following Cont et al. (2014) and Silantsev (2019), bar mid-quote change is regressed on contemporaneous and one-bar lagged OFI:

$$\Delta p_t = \alpha + \beta_0 O_t + \beta_1 O_{t-1} + \varepsilon_t, \quad (1)$$

where $\Delta p_t = m_t - m_{t-1}$ is the change in the mid quote between bar boundaries and O_t is the bar OFI, so β_0 is in dollars of mid-quote change per contract of net imbalance. Standard errors are Newey-West HAC (Newey and West, 1987) with five lags. Table 1 reports the estimates on the full panel, the 2021-02 to 2023-12 in-sample (IS) window, and the 2024-01 to 2026-08 out-of-sample (OOS) window.

Window	n (bars)	$\hat{\beta}_0$	HAC $t(\beta_0)$	$\hat{\beta}_1$	HAC $t(\beta_1)$	R^2
Full sample	41,055	+0.247	+70.2	-0.027	-14.4	0.29
In-sample 2021–23	12,344	+0.300	+44.9	-0.025	-7.0	0.31
Out-of-sample 2024–26	28,711	+0.215	+58.1	-0.026	-13.3	0.30

Table 1: OLS, Newey-West HAC (five lags). $p < 10^{-6}$ on both coefficients in every window.

Figure 1 shows the out-of-sample relation. The IS and OOS R^2 are nearly identical (0.31 and 0.30), while the slope itself is less stable, moving from +0.300 in-sample to +0.215 out-of-sample. Both R^2 figures are below the two-thirds level of Cont et al. (2014) for liquid US equities, albeit at a different sampling frequency; Section 4 repeats the comparison at their ten-second interval.

This regression is contemporaneous: it measures how the mid quote moves together with the flow inside a bar (median OOS bar duration ≈ 17 minutes), not whether flow forecasts the next bar. Regressing the next bar’s mid-quote change on lagged OFI alone gives $R^2 < 0.001$ out of sample, so the content of Table 1 is price impact rather than forecastability. Whether the flow carries predictive content at horizons inside the bar is a question the bar regression cannot resolve.

4 Robustness to the bar scale

The ten-bars-per-day calibration is a convention rather than a fitted parameter, and the result does not depend on it. Rebuilding the panel at every target from one to one thousand bars per day, with the threshold recomputed on the in-sample window at each scale, leaves the out-of-sample R^2 between 0.29 and 0.32 everywhere, while $\hat{\beta}_0$ rises from +0.17 on the coarsest bars

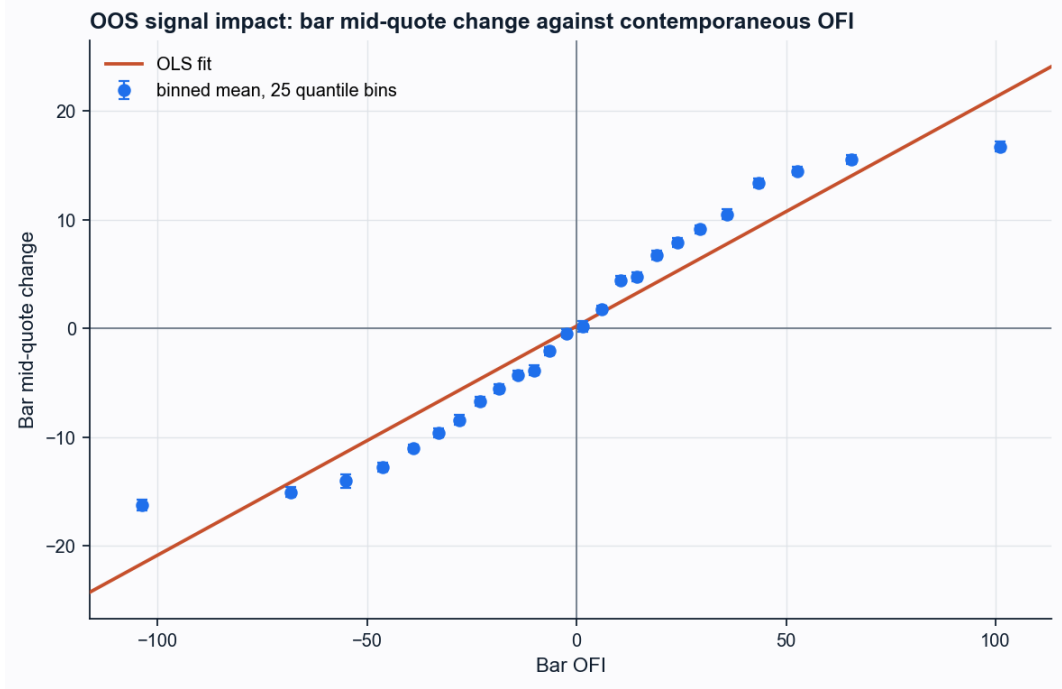


Figure 1: Out-of-sample bar mid-quote change against contemporaneous OFI, 25 quantile bins with ± 1 standard error and the least-squares fit of Δp_t on O_t alone. The relationship is monotonic, with impact per unit of flow attenuating in the tails: both extreme bins lie below the fitted line in absolute value. Full estimates are in Table 1.

to $+0.25$ on the finest (Figure 2). Median durations run from almost four hours down to the panel’s one-minute resolution, which binds beyond roughly two hundred bars per day: past that point most bars complete within a single minute, so the finest scales are effectively one-minute time bars. No bar size in this range would change the conclusions of Table 1.

The one-minute floor is a property of the panel, not the tape, so the sweep is repeated on a ten-second event grid rebuilt from the raw TBBO stream for the out-of-sample window; the rebuilt buckets aggregate back to the minute panel exactly. Dollar-volume bars on this grid, from ten to five thousand bars per day, hold the out-of-sample R^2 between 0.31 and 0.32, and on plain ten-second intervals, the sampling grid of Cont et al. (2014), the regression gives $\hat{\beta}_0 = +0.26$ ($t = 189$, $R^2 = 0.32$) on 1.6 million trade-active buckets. The gap to the equity two-thirds is therefore not an artifact of bar size: at matching intervals this contract’s flow explains about half as much of the mid-quote variation as Cont et al. report for liquid US equities, with the caveat that TBBO observes the book only at trade instants. The lagged coefficient shrinks steadily as bars get finer and is indistinguishable from zero on the ten-second grid ($t = -0.3$), so the small negative $\hat{\beta}_1$ of Table 1 reflects netting within coarse bars rather than a persistent reversal in the flow.

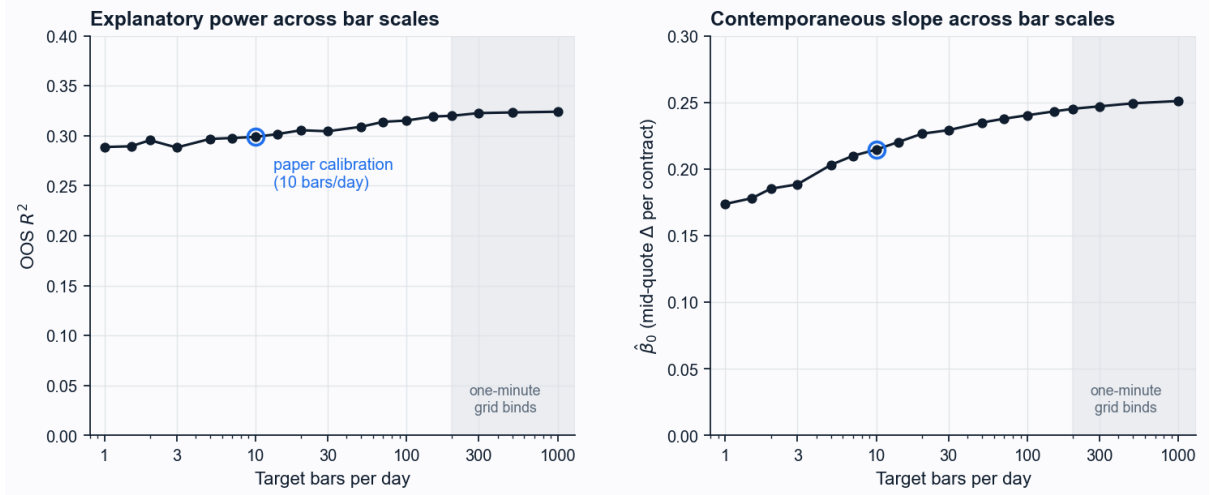


Figure 2: Out-of-sample R^2 (left) and contemporaneous slope $\hat{\beta}_0$ (right) as the bar threshold is varied from one to one thousand target bars per day, recalibrated on the in-sample window at each scale. The circled point is the ten-bars-per-day calibration used throughout the paper; in the shaded region the one-minute grid binds and most bars are single minutes.

5 Conclusion

The contemporaneous price impact of OFI is strong on CME Ether futures, and its explanatory power carries out of sample essentially unchanged, with the slope large and precisely estimated throughout. The result describes how this market absorbs order flow; it does not supply a trading signal.

The tape bounds what can be measured. TBBO is trade-sampled, so the book is observed only when a trade prints, and it carries no depth beyond the touch. A quote-level feed would permit measuring the relation at horizons inside the bar, where any predictive content would live, and a proper treatment of execution; these are the natural extensions.

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